

Impact of grid-forming wind turbine model order reduction on interaction studies with inter-area oscillations[☆]

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ABSTRACT

The energy transition towards carbon-neutrality is rapidly changing power systems as more wind energy is increasingly being integrated. As the share of power electronics interfaced energy resources increases, stability concerns arise due to the devices' distinct behavior with respect to traditional equipment. In this context, grid forming converter control (GFM) is highlighted as a system stabilizing solution. Especially, wind turbines (WT) with GFM functionalities are a potential technology to support system stability while providing renewable energy. However, the impact of GFM WT on traditional power stability phenomena related to synchronous generators (SG) requires further investigation. This paper conducts an analysis of the impact of GFM-WT modeling fidelity for conducting power system studies efficiently. In addition, we focus on inter-area modes and their interplay with low frequency turbine modes linked to the drive-train and GFM control. The validity and effectiveness of the proposed reduced-order models for small-signal stability analysis are demonstrated by frequency response comparison and an investigation of the model structure from a state-space perspective. Lastly, the impact of different GFM-WT drive-train and control parameters on inter-area oscillations is studied for a modified two-area power system. All the results discussed in this paper are validated through electromagnetic transient (EMT) simulation.

1. Introduction

Wind power technology is one of the key technologies to achieve carbon-neutrality, and many countries are planning massive integration of wind energy [1]. However, renewable energy sources are usually integrated into power systems via power electronics converters and controls which impose new stability challenges [2].

To address these dynamic stability challenges, modeling fidelity is important to sufficiently ensure the accuracy of analysis. An attempt has been made to provide generic models for assessing stability with electromagnetic transient (EMT) simulation [3]. This is an effective way of investigating system stability in terms of accuracy with the capability to account nonlinearity and complex controls. However, EMT-oriented time-domain simulation investigations are extremely time consuming and usually do not provide useful insights about the root-cause of dynamic issues.

To gain a deeper understanding of the system dynamics, which can be utilized to prevent instability problems and guide system design, modal analysis has been widely utilized [4]. Modal analysis typically requires an analytical model which is then linearized into a state-space form to enable eigenvalue and participation factor (PF) analysis, which reveals the dominant causes and characteristics of the system's oscillatory modes.

Even though modal analysis via PF provides valuable results, it requires detailed models to capture the entire system dynamics. The large model size and associated complexity at times hinders an efficient analysis. Therefore, reduced order models capturing only the characteristics relevant to specific phenomena under study are often adopted. For example, the reduced order model of the network and synchronous generators (SG), where the electromagnetic dynamics are neglected, are commonly utilized to investigate inter-area oscillations [5].

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While these stability analysis techniques have been widely adopted, stability concerns due to the interaction between power converters and other systems have been growing with the increased penetration of renewable energy. To address these stability concerns, the grid forming strategy is highlighted due to its grid-supportive characteristics [6].

Grid forming control (GFM) is often designed to emulate synchronous generator's inherent supportive characteristics with their slow internal electromechanical dynamics [7]. There are several possible strategies for grid-forming control schemes for wind turbines (WT) [8]. However, due to these slow dynamic characteristics, the GFM control strategy is rarely compatible with the DC voltage control needed in the back-to-back configuration of type 4 wind turbines (WT) as it requires comparatively fast power control dynamics to regulate the DC voltage [9]. Therefore, the DC voltage control can be performed by the rotor side converter (RSC) of the WT, and the GFM control is performed by the grid side converter (GSC).

However, although this control configuration enables grid supportive characteristics with grid forming wind turbines (GFM WT), the interaction between the turbine and network under perturbations is of concern. This is addressed with the implementation of a damping controller for drive train oscillations in Ref. [10,11]. Although these previous works adopt a reduced order model for the RSC, the justification for said model selection is not sufficient and an adequate rationale is needed to better understand the limitations and scope of said reduced order models. Therefore, this paper proposes a novel internal structure analysis method utilized for justifying the reduced order models. In addition, the impact of the modeling fidelity will be comprehensively investigated by comparing the detailed and reduced models. These analyses show that reduced order models sufficiently reflect the dynamics of GFM WT for system-wide low-frequency interaction studies.

Next, interaction assessment studies are conducted between GFM WT and SG dynamics in power systems, the so-called inter-area oscillations [12]. For this purpose, the derived reduced order GFM WT models are used to systematically investigate the impact of GFM WT on the inter-area modes via eigenvalue trajectory calculation and frequency response under different parameters. The results show that GFM control significantly influences the stability of inter-area modes. In addition, results show that drive-train characteristics also significantly influence the interaction characteristics for GFM WT and between the interconnected system and GFM WT. Lastly, the analysis results are validated through nonlinear time-domain simulations using PSCAD/EMTDC.

2. System modeling and description

In this section, the system modeling approach adopted in this paper is introduced. The approach is subsequently applied in next sections for inter-area oscillation investigations and model order reduction of GFM WT.

2.1. Synchronous generator model for inter-area oscillation studies

To study the impact of GFM WT on inter-area oscillations, a detailed synchronous generator model is adopted [4]. The machine is modeled in the dq -frame including one damper winding and the field circuit in the d -axis, two damper windings in the q -axis and an equivalent lumped-mass equation of motion. In addition, it is equipped with standard IEEE automatic voltage regulator (AVR) and turbine-governor (GOV) models, AC Exciter AC4C and modified steam turbine-governor model IEEE1, respectively. This detailed SG-based power plant model incorporating electromagnetic, mechanical, thermal and control dynamics is used to analyze inter-area oscillation in Section 4. More details about this SG modeling can be found in A. Throughout the paper $\mathbf{p}$ denotes the time-derivative operator and other bold symbols to denote vectors and matrices.

2.2. Detailed model of GFM WT

Several approaches to the modeling of GFM WT based on multiple GFM control methods exist, such as virtual synchronous machine (VSM) and droop-based strategy [13]. As they have very similar dynamic characteristics and responses in the electromechanical range, VSM control is adopted in this paper [14]. Compared to grid following control (GFL) WT, it also has freedom for assigning the DC voltage regulating station, i.e., RSC, or GSC. The impact of selecting the control station is broadly discussed in Ref. [9]. In this paper, RSC regulates the DC voltage as the desired bandwidth of DC voltage regulation is usually higher than that of the GFM active power control, and thus avoids incompatible functions. However, there is a lack of detailed investigation of the reduced order model's validity and impacts. Therefore, this paper investigates it in depth via various approaches, i.e., state-space and frequency domain methods, by comparing detailed and reduced-order models. In addition, the proposed internal structure analysis further supports the understanding the impact of model order reduction on the interaction studies.

The power extracted from the wind by the turbine is mainly dependent on wind velocity V_w , which varies according to the geographical environment. The WTs under study are operated with GFM control and maximum available power point tracking (MPPT) algorithm during normal operating conditions [15]. The active power captured by the rotor P_w varies depending on the rotor speed Ω_r and wind velocity V_w with the power efficiency C_p which can be determined by the tip speed ratio (TSR) Γ and pitch angle β described as follows:

$$\begin{cases} P_w = \frac{1}{2} \rho \pi R^2 C_p V_w^3 = k_o C_p V_w^3 & \& \Gamma = \frac{R\Omega_r}{V_w} \\ C_p(\Gamma, \beta) = c_1 \left(\frac{c_2}{\Gamma_k} - c_3\beta - c_4 \right) e^{\frac{-c_5}{\Gamma_k}} + c_6\Gamma \\ \frac{1}{\Gamma_k} = \frac{1}{\Gamma + c_7\beta} - \frac{c_8}{\beta^3 + 1} \end{cases} \quad (1)$$

where R is the rotor blade radius, ρ is the air density, k_o is a rotor related constant, and non-dimensional rotor efficiency coefficients c_1 - c_8 are as follows: $c_1 = 0.5176$, $c_2 = 116$, $c_3 = 0.4$, $c_4 = 5$, $c_5 = 21$, $c_6 = 0.0068$, $c_7 = 0.08$, $c_8 = 0.035$. In addition, it is assumed that the wind turbine is operated with an optimal pitch angle during the MPPT region, i.e., $\beta = 0$, so as to focus on its under-rated operation, which is the dominant operation regime for WTs. More details about this wind turbine model can be found in Ref. [15]. The GFM function performed by the GSC is expressed as:

$$\mathbf{p}\omega_{wt} = \frac{P_{wt}^* - P_{wt} + P_{d,wt} + P_{k,wt}}{2H_{vsm}} \quad (2)$$

$$\theta_{wt} = \frac{\omega_{wt}}{s}, \quad \begin{cases} P_{k,wt} = K_{vsm}(\omega_{wt}^* - \omega_{wt}) \\ P_{d,wt} = D_{vsm}(\omega_{ac} - \omega_{wt}) \end{cases} \quad (3)$$

where P_{wt}^* and P_{wt} are MPPT reference and power output, respectively, H_{vsm} and ω_{wt} are the virtual inertia of GFM control and angular speed for the controller, respectively, K_{vsm} & D_{vsm} are the droop and damping constants, and $P_{d,wt}$ & $P_{k,wt}$ are the damping and droop power of the WT, respectively. The angle θ_{wt} in rad is generated from the GFM control and is used in combination with the reactive power control, virtual impedance and current control's outputs to generate the modulated voltage signals $\mathbf{u}_{wt}$. Furthermore, the grid side frequency ω_{ac} is estimated via a Phase-Locked Loop [16]. This estimated ω_{ac} is solely used for the damping of the VSM control loop. Note, this is contrary to its use for synchronization in GFL WTs.

The wind turbine model description is presented in Fig. 1. A standard two-mass drive train model can be described by Eqs. (4)–(6)

$$\mathbf{p}\Omega_r = \frac{T_r - K_t(\theta_r - \theta_{g,wt}) + D_t(\Omega_r - \Omega_{g,wt})}{2H_r} \quad (4)$$

$$\mathbf{p}\Omega_{g,wt} = \frac{K_t(\theta_r - \theta_{g,wt}) + D_t(\Omega_r - \Omega_{g,wt}) - T_{g,wt}}{2H_{g,wt}} \quad (5)$$

$$\mathbf{p}\theta_t = \Omega_r - \Omega_{g,wt}, \quad \theta_t = \theta_r - \theta_{g,wt} \quad (6)$$

where T_r and $T_{g,wt}$ are the rotor and generator torques, respectively as $T_r = P_w/\Omega_r$, and generator's power $P_{g,wt} = T_{g,wt}\Omega_{g,wt} \cdot K_t$ and D_l are the spring and damping constants of the drive train, $\Omega_{g,wt}$ is the angular velocity of generator, θ_r & $\theta_{g,wt}$ are the rotor and generator angles, and θ is the angular position difference between the rotor and the generator. The electrical machine of the GFM WT is a commonly used permanent magnet synchronous generator (PMSG), which has similar dynamics to SG in A with some simplifications, so it can be expressed as

$$u_g^d = r_g i_r^d + \mathbf{p} \psi_g^d - \Omega_{g,wt} \psi_g^q \quad (7)$$

$$u_g^q = r_g i_r^q + \mathbf{p} \psi_g^q + \Omega_{g,wt} (\psi_g^d + \psi_M) \quad (8)$$

$$T_{g,wt} = i_r^q \psi_M - i_r^q i_r^d (L_g^d - L_g^q) \quad (9)$$

where $\mathbf{u}_g = [u_g^d, u_g^q]^T$ are the PMSG voltages, r_g and $\mathbf{L}_g = [L_g^d, L_g^q]^T$ are the PMSG winding resistance and inductances, $\boldsymbol{\psi}_g = [\psi_g^d, \psi_g^q]^T$ are flux linkages, ψ_M is the permanent magnet rotor flux, $\mathbf{i}_r = [i_r^d, i_r^q]^T$ is the PMSG's current vector, which represents the RSC's AC current. $\omega_{g,wt}$ and $T_{g,wt}$ are the angular speed and torque, respectively. The RSC side model and the DC bus dynamics and control can be expressed as

$$\mathbf{p}u_{dc} = \frac{P_{rsc} - P_{wt}}{u_{dc}C_{dc}} \quad \& \quad i_r^{q,*} = G_{dc}e_{dc} + \tilde{i}_f \quad (10)$$

$$i_f = \frac{\tilde{i}}{1 + sT_g} \quad \& \quad \tilde{i} = P_{wt}/u_{dc} \quad (11)$$

$$u_g^d = G_{i,r}^d (i_r^{*,d} - i_r^d) - \omega_{g,wI} \psi_g^q \quad (12)$$

$$u_g^q = G_{i,r}^q(i_r^{*,q} - i_r^q) + \omega_{g,wt}(\psi_g^d + \psi_M) \quad (13)$$

where u_{dc} and C_{dc} are the DC bus voltage and capacitor of the GFM WT. P_{rsc} denotes the RSC side active power into RSC. G_{dc} denotes the DC voltage control transfer function (PI controller), respectively. In addition, $\tilde{i}$ and $\tilde{i}_f$ are the feed-forward current and its measured value. This feed-forward current is the GSC side current to improve the controllability over the DC voltage [9]. Here, the losses across the converter stations and its DC bus are assumed to be negligible. T_g is the feed-forward current low-pass filter (LPF) time constant, $G_{i,r}^d$ and $G_{i,r}^q$ are the current controller transfer functions (PI controllers), and i_r^d and i_r^q are dq -frame current reference signals.

In addition to Eq. (2) and Eq. (3), according to the desired GFM behavior of a voltage source behind impedance, the GSC model has a quasi-static virtual impedance (VI) loop cascaded with a slower reactive power controller, which are presented in Fig. 2. The virtual resistance and inductance are denoted by r_v and l_v , respectively, which result in a virtual admittance magnitude of $Y_v = 1 / \sqrt{r_v^2 + l_v^2}$. The reactive power reference, Q_{ref}^* , is set to 0 in all study cases. More details about this GFM and GSC control approach can be found in Appendix B [17]. The adopted parameters are shown in Appendix C.

This detailed GFM WT model's validation is carried out by comparing the terminal dynamics at the grid-connection point of the linearized state-space analytical model with respect to the frequency scan results of its non-linear EMT implementation. This frequency scanning method enables the comparison of the frequency responses between the analytical model and the nonlinear EMT model to validate the constructed model; thus, a matched analytical model can be used for accurate stability analysis. The detailed information about the frequency scan of EMT models can be found in Ref. [18]. This response of the analytical model can be obtained from the model with a given state-space model $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$ for its state, input, output, and feed-through matrices, respectively. Together with the states $\mathbf{x}(t)$, inputs $\mathbf{u}(t)$, outputs $\mathbf{y}(t)$ as

$$\dot{\mathbf{x}}(t) = \mathbf{A} \mathbf{x}(t) + \mathbf{B} u(t) \quad (14)$$

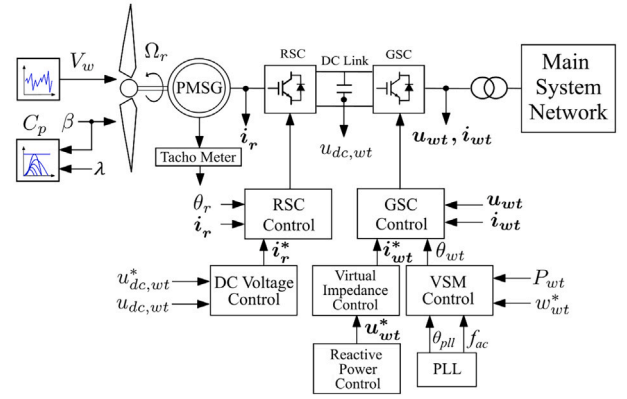


Fig. 1. GFM WT model and controls.

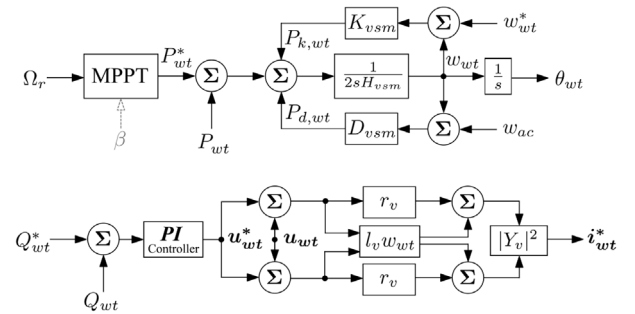


Fig. 2. GSC VSM control scheme of the GFM WT.

$$\mathbf{y}(t) = \mathbf{C} \mathbf{x}(t) + \mathbf{D} \mathbf{u}(t) \quad (15)$$

$$G(s) = C(sI - A)^{-1}B + D \quad (16)$$

where $G(s)$ denotes the transfer function of the model, which is admittance $Y_{dq}(j\omega)$ in this case by choosing dq currents. Here, wind velocity is also considered as its input. For the output, dq voltages are used.

The results are shown in Fig. 3. The constructed state-space analytical model (solid lines) matches well with the nonlinear model in EMT simulation (blue circles). Compared to GFL WT's frequency responses, which can be found in Appendix, the GFM WT presents a drive-train oscillatory mode around $10^{0.33} \approx 2.1$ Hz, which is especially obvious in the frequency response of $|Y_{dd}|$, i.e., there is a chance of interaction between this drive-train mode and the hosting electrical network. This oscillation mode frequency $\omega_{l,osc}$ and its damping ratio ζ_l can be estimated from the equivalent physical parameters as [19].

$$\omega_{t,osc} = \sqrt{\frac{K_t(H_r + H_{g,wl})}{2H_r H_{g,wl}}}, \quad \zeta_t = \frac{D_t(H_r + H_{g,wl})}{4H_r H_{g,wl} \omega_{t,osc}} \quad (17)$$

which appears as a peak in the frequency response of the admittance. It can be further confirmed that this is related to the drive-train via PF analysis of the closest frequency mode $\lambda = -0.5 \pm j14.05$ in Fig. 4 with a dominant participation of drive-train states.

2.3. Reduced order model of GFM WT

Depending on the study purpose or the frequency region of interest in dynamics, some of the dynamics in the model might be simplified or neglected. For example, RMS studies neglect fast electromagnetic dynamics for efficient analysis of very low frequency dynamics studies (<2 Hz). In contrast, EMT studies with more detailed models enable accurate dynamics studies across wider frequency ranges, including sub-synchronous phenomena and high frequency interaction studies [20].

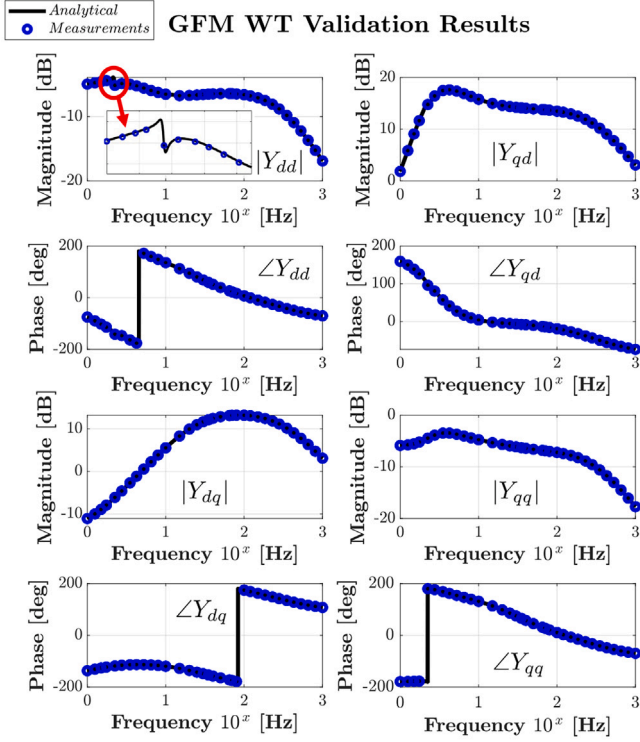


Fig. 3. Frequency response validation of detailed GFM WT model between the analytical model and nonlinear EMT model.

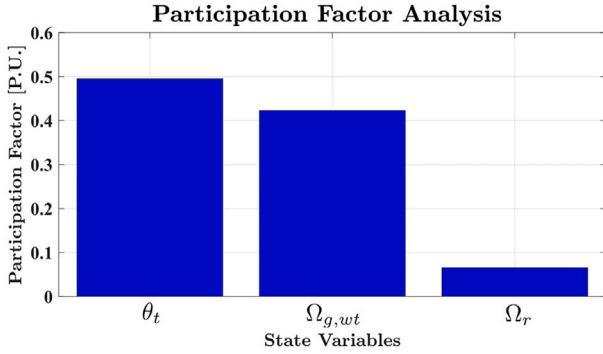


Fig. 4. Participation factors of the mode $\lambda = -0.5 \pm j14.05$.

Therefore, it is customary to carefully investigate the impacts of model order reduction as selected key dynamics are still needed to enable accurate interaction studies on large-scale systems. To compromise on the trade-off between the complexities and computational burdens arising from large-size models of GFM WT and the accuracy of stability analysis, a reduced order model is proposed in Ref. [9]. However, the validity of the reduced order model is not discussed, which might lead to incorrect stability conclusions under different conditions. To fill this gap, we compare different reduced-order models with respect to the detailed model. The accuracy of the reduced order models and the impact of the model order are investigated based on both modal analysis with eigenvalue analysis and frequency-response examinations.

The starting premise is that major model order reduction can be achieved by simplifying the rotor-side dynamics. Furthermore, the RSC's current controller is designed to respond much faster than the time scales of interest in this paper, which are in the low-frequency range, its detailed dynamics can be disregarded in Eq. (12) and Eq. (13). In addition, the current-related machine dynamics with Eq. (7) and Eq.

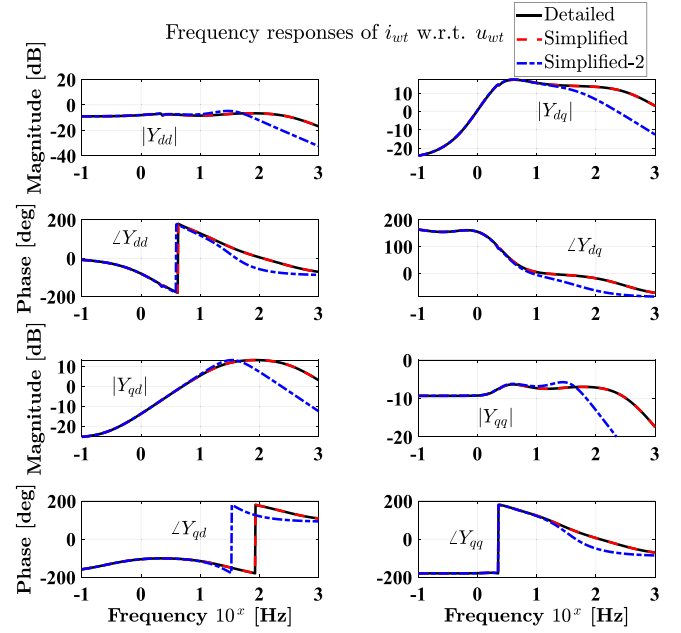


Fig. 5. Frequency response comparison between detailed and simplified GFM WT models with respect to grid side voltages.

(8) are also simplified under the assumption that the RSC tightly and quickly regulates them. Here, the simplification is achieved by replacing the associated differential equations and state variables with simple algebraic equations. In contrast, the drive-train dynamics are crucial concerns of GFM WT, therefore they are retained in the reduced models. Now, the state variables of rotor side dynamics, i_r and $\varphi_{i,r} = [\varphi_{i,r}^d, \varphi_{i,r}^q]^T$, are replaced by the rotor-side active power P_g as a new state variable. Therefore, now the total number of state variables is 16 (as four variables ($i_r^d, i_r^q, \varphi_{i,r}^d, \varphi_{i,r}^q$) are replaced with one new variable P_g). This reduced order model is referred to as *Simplified-1* or just *Simplified* in this paper. This rotor-side dynamics of this simplified model are then described by P_g as

$$P_g = \frac{(i_r^{q,*} + \tilde{i}_f)u_{dc}}{1 + sT_c} \quad (18)$$

where the time constant T_c is used to approximate the rotor-side current dynamics as a first-order lag process. By assuming ideal rotor side current control operation for model order reduction, this time constant is determined based on the current controller bandwidth. Here, the underlying assumptions are perfect modulation and current regulation; thus, the produced power is solely controlled by q -axis current ($i_r^d = 0$). In addition, the losses across the RSC and PMSG are neglected. The validity of these assumptions and premises is confirmed from the frequency response comparison in Fig. 5 between the *Detailed* and this *Simplified-1* models. Therefore, it can be concluded that this reduced model is adequate for a wide range of interaction studies as the frequency response closely matches in the evaluated frequency range. Therefore, this model can be used to study the interactions when interconnected to a larger system, as these interactions are driven by the characteristics of the WT terminal dynamics captured in its admittance and the rest of the system. However, it is worth noting that this reduced order model may not be valid for its device-level stability studies, i.e., those analyses focusing on internal subsystem interactions within the WT rather than interaction with other systems through electrical interconnections.

Moreover, further model order reduction can be carried out in the GSC, which is referred to hereafter as *Simplified-2*, to investigate the validity of further model order reduction by neglecting current control dynamics in its GSC. This is based on the same reasons as the

Table 1
List of the resultant state variables for each model.

Model type	Model order	State variables
Detailed	19-th	$i_{wt}^d, i_{wt}^q, \varphi_i^d, \varphi_i^q, \varphi_{pll}, \theta_{pll}, \omega_{wt}, \theta_{wt}, \varphi_q, u_{dc}, \tilde{i}_f, \varphi_{dc}, \Omega_{g,wt}, \Omega_r, \theta_i, \varphi_{fr}^d, \varphi_{fr}^q, i_r^d, i_r^q$
Simplified-1	16-th	$i_{wt}^d, i_{wt}^q, \varphi_i^d, \varphi_i^q, \varphi_{pll}, \theta_{pll}, \omega_{wt}, \theta_{wt}, \varphi_q, u_{dc}, \tilde{i}_f, \varphi_{dc}, \Omega_{g,wt}, \Omega_r, \theta_i, P_g$
Simplified-2	14-th	$i_{simp}^d, i_{simp}^q, \varphi_{pll}, \theta_{pll}, \omega_{wt}, \theta_{wt}, \varphi_q, u_{dc}, \tilde{i}_f, \varphi_{dc}, \Omega_{g,wt}, \Omega_r, \theta_i, P_g$

case for *Simplified-1*, i.e. the current control dynamics are much faster than target interactions in the relatively lower frequency range below approximately 10 Hz.

$$\mathbf{p}_{simp}^d = \frac{1}{T_{cc}}(i_{wt}^{d,*} - i_{simp}^d) \quad (19)$$

$$\mathbf{p}_{simp}^q = \frac{1}{T_{cc}}(i_{wt}^{q,*} - i_{simp}^q) \quad (20)$$

The states $(i_{wt}^d, i_{wt}^q, \varphi_i^d, \varphi_i^q)$ can be replaced by i_{simp}^d, i_{simp}^q , which represent the designed closed-loop current control dynamics as expressed in (19)-(20) with time constant T_{cc} , and $i_{wt}^{d,*}, i_{wt}^{q,*}$ being the reference currents from the other VSC controls. T_{cc} can be determined with same approach as in T_c . As a result, the total number of state variables for *Simplified-2* is 14 reduced from 16 in *Simplified-1*. In addition, the summarized list of state variables for each type of model is provided in Table 1.

To verify the validity of previous simplifications and resulting reduced order models, a frequency response comparison is carried out in Fig. 5 including *Simplified-2* case. It shows that both the *Simplified-1* model, in red dashed lines, and the *Simplified-2* model, in dash-dotted blue lines, very well capture the overall GFM WT dynamics in low frequency region compared to the detailed model, in solid black lines, especially the relevant drive-train dynamics of crucial concern in GFM WTs. Fig. 5 suggests that there is not a significant difference between these models as seen from the network up to 10 Hz, i.e. as seen from the terminal voltages and currents. Therefore, these reduced order models are useful to investigate the stability after interconnecting the GFM WT with the rest of the power system. For the case of *Simplified-2*, even though it captures the lower frequency dynamics properly, including drive-train oscillation mode, it shows a significant difference in the high-frequency region. This means that this type of reduced order model is not appropriate for high-frequency interaction studies, such as the excitation of network resonances or any network-side disturbances above the second harmonic.

In addition to its electrical terminal voltages, the wind velocity V_w is also an external input to the GFM WT, as opposed to the commonly considered simplification of VSCs with an ideal DC side or HVDC connection. This aspect is addressed in Section 3. It is important to ensure that the reduced order model can also capture the impact of varying wind velocity, as wind fluctuations can translate to the WT output and might impact system stability when containing significant frequency content close to existing system modes [21].

3. Comparison study of modeling fidelity

To ensure a sufficiently accurate stability analysis, it is important to guarantee the fidelity the employed dynamic models. If model order reduction is performed, depending on the approach, the model accuracy in certain frequency regions and/or with respect to certain inputs/outputs can be degraded and may lead to incorrect stability conclusions. In this section, in addition to previous network-side response validation, the fidelity of various reduced order models is assessed by means of eigenvalue comparison with modal analysis, frequency response comparison, and internal structure analysis.

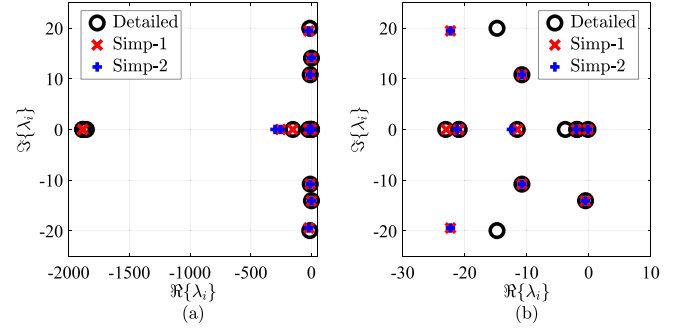


Fig. 6. Eigenvalue comparison between detailed and simplified GFM WT models. (a) Eigenvalues with lower frequencies. (b) Zoom into the lower damping region.

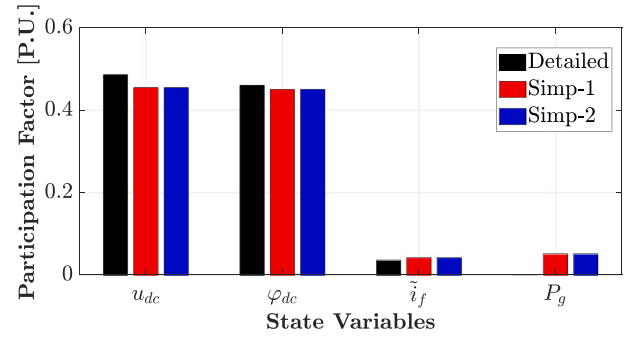


Fig. 7. Comparisons of PFs about the DC voltage mode $\lambda = -17 \pm j21$ of *Detailed* and $\lambda = -22 \pm j19$ of *Simplified-1* and *Simplified-2*.

3.1. Eigenvalue comparison study

To check the validity of the reduced order models and their impact on the system dynamics, the GFM WT eigenvalues λ are compared between the detailed and the reduced order models in Fig. 6. The results show that most low-frequency eigenvalues are largely matched except for one conjugate pole pair of the simplified model. This shows that the model order reduction process could also introduce fictitious eigenvalues as a result of approximating high-frequency dynamics by slower dynamics and algebraic equations or by introducing new states.

To get more insights from these mismatches in the eigenvalues, PF analysis is performed for the three models on the mode with $\Im\{\lambda\} \approx 20$ rad/s and presented in Fig. 7. All the participating dynamics of the concerned eigenvalue are rotor-side, with differences arising from model order reductions. Since *Simplified-1* and *Simplified-2* yield identical results, it can be confirmed that this difference comes from the model order reduction from *Detailed* to *Simplified-1*. However, this mode difference does not affect the interaction studies between sub-systems, as their interaction behaviors can be seen from the admittances, which remain identical between *Detailed* and *Simplified-1* in Fig. 5. Instead, it is primarily related to its internal stability. Thus, this aspect is more about the device-level design, focusing on internal stability rather than on interaction studies. Consequently, the results of terminal dynamic behavior are not significantly affected, provided that internal stability is ensured and the reduced-order models can be effectively used for studying low-frequency interactions.

In addition, Fig. 7 shows that the reduced order models are affected by the artificially introduced state P_g . While introducing P_g , the impact of DC voltage dynamics is slightly weakened, which explains the minor increase in damping with respect to the detailed model. Therefore, this model order reduction strategy should be carefully applied when the DC voltage dynamics are the focus of the studies. The eigenvalue approach for validating reduced-order models provides useful results

for understanding the impact of different model aspects on specific modes.

To further understand the impact of differences in the DC voltage mode from the previous subsection, observability can be investigated for each model case to see the impacts of different modes. The observability $o_{i,k}$ can be calculated for a specific k -th mode with a specific i -th output.

$$o_{i,k} \triangleq |(Cv_k)_i| = |(C_i v_k)| \quad (21)$$

where v_k is k -th right eigenvector and C is output matrix of state space model and C_i is its i -th row. Here, the concerned input and output of models are input voltage vectors and output current vectors to focus on the interaction with other subsystems rather than GFM WT itself. The observability of the concerned DC voltage mode is almost zero for all types of models as $[6.34\text{e-}04, 1.806\text{e-}05]$, $[4.285\text{e-}04, 1.542\text{e-}05]$, and $[4.628\text{e-}04, 1.555\text{e-}05]$ with respect to dq output currents for *Detailed*, *Simplified-1*, and *Simplified-2*, respectively. It implies that one cannot see the impact of this mode in the output currents of GFM WT. Therefore, these DC voltage mode-related impacts cannot be seen from its grid connection sides. It reconfirms the matched results of frequency responses between the *Detailed* and *Simplified-1* models in Fig. 5; thus, there is no mismatch between these frequency responses even though they have differences in the DC voltage-related eigenvalue.

3.2. Frequency response analysis

Subsequently, frequency response comparisons are further carried out to gain a deeper understanding. To validate the accuracy of the reduced-order model of GFM WT and its impact on the study of interactions with the interconnected system, the frequency responses of each model with respect to relevant input/output signals are further compared.

The previous section introduced the comparison of the output current i_{wt} response with respect to the network-side dq -axis voltage u_{wt} , i.e., the GFM WT admittance matrix, which is presented in Fig. 5. The comparison results indicate that there are no significant differences between the detailed model and the Simplified-1 model. However, the further simplified model, Simplified-2, shows a significant mismatch in high-frequency regions above approximately 10 Hz as expected; thus the presented reduced order models are only admissible for low frequency interaction studies.

In addition, compared to a simple VSC study case model, the WT model has one more input, i.e., the wind speed V_w , which evidently can also excite power system dynamics. Even though the previous results are widely used for frequency domain-based stability studies, the effect of primary resource-driven perturbations, such as wind gusts, have been neglected so far. However, in order to accommodate high wind energy shares through the GFM WT, there is a need to better capture and understand the impact of the primary resource dynamics on the devices and the rest of the power system. To fill this gap, frequency responses of i_{wt} with respect to V_w are extracted and shown in Fig. 8. The responses roughly depict an attenuation shape which effect increases over the frequency. Mostly the d -axis current is affected by the wind speed at low frequencies due to its direct relation to active power as the reference frame is aligned with the WT terminal voltage at steady-state. These results also show a similar good match with the detailed model as in Fig. 5. Similarly, mismatches in the high-frequency region occur when simplifying the dynamics of the RSC current controller in the simplified model, and it is worse in Simplified-2 with further skipping of GSC current controller states. These results imply that the accuracy of the frequency response will unavoidably be lost after performing model order reduction. Therefore, model order reduction should be carefully carried out depending on the study purpose and frequency region of interest to preserve the accuracy of the relevant dynamics, which is a typical challenge of model order reduction problems. As this paper focuses on inter-area oscillations which are system-wide phenomena with very low

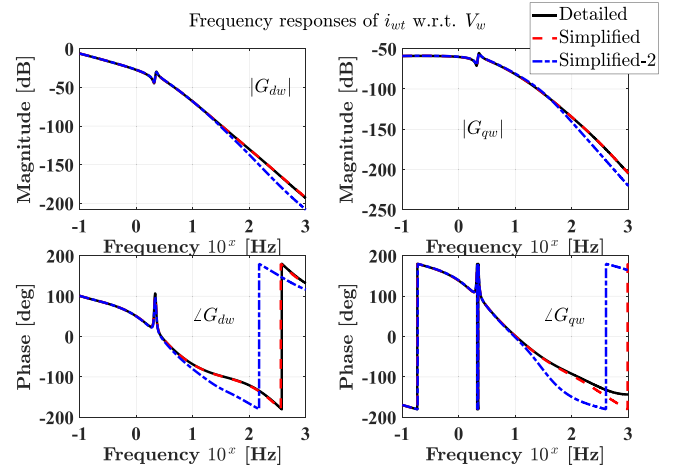


Fig. 8. Frequency response comparison between detailed and simplified GFM WT models with respect to wind velocity input.

frequency around 0.1 – 1 Hz, the investigated reduced order models sufficiently preserve their fidelity up to one decade above the inter-area dynamics and thus they are considered adequate for such stability studies.

Also, the modal analyses suggest that it might not be sufficient to only compare the eigenvalues, which is the most commonly used method by previous works for validating reduced order models. Here, it would be helpful to provide more supporting reasons for a better understanding of the causal impact of model order reduction. Consequently, the next section studies the internal model structure so as to gain more insights into the loss of coupling channels and dynamics in the different models.

3.3. Internal structure analysis

To overcome the limited understanding that can be extracted from the modal analysis comparison of the different models, this paper proposes a new approach to gaining better insights into the impacts of model order reduction. While frequency responses effectively show the model's validity for studying phenomena around specific frequency ranges, these do not provide a clear relationship between the retained and neglected dynamics. These gaps in eigenvalue and frequency domain analyses could be supplemented by looking at the model's internal structure as captured by its state matrix. This matrix links the system states and their derivatives, which define the system dynamic characteristics such as natural frequencies, damping and participating variables. The internal structures of each model are presented in Fig. 9. The results show which state variables are coupled with which state derivatives. Here, the state variables x for detailed model are $x_{detail} = [i_{wt}^d, i_{wt}^q, \phi_i^d, \phi_i^q, \theta_{pll}, \theta_{pll}, \omega_{vsm}, \theta_{vsm}, \phi_q, u_{dc}, \tilde{f}, \phi_{dc}, \Omega_{g,wt}, \Omega_r, \theta_t, \phi_{i,r}^d, \phi_{i,r}^q, i_r^d, i_r^q]^T \in \mathbb{R}^{19}$. The state vector of the simplified model is $x_{simp} \in \mathbb{R}^{16}$, which omits $\phi_{i,r}^d, \phi_{i,r}^q, i_r^d, i_r^q$ from x_{detail} , and instead it has a lumped state P_g . The states of the further simplified model, Simplified-2, are $x_{simp,2} \in \mathbb{R}^{14}$, which result from neglecting $i_{wt}^d, i_{wt}^q, \phi_i^d, \phi_i^q, \phi_{i,r}^d, \phi_{i,r}^q, i_r^d, i_r^q$ from x_{detail} , and instead, it has a simplified representation by i_{simp}^d, i_{simp}^q . The GSC and RSC state variables are enclosed by a colored box in orange and burgundy, respectively. These structural results reveal that both RSC and GSC dynamics are coupled through the 10th state, highlighted by the magenta dashed lines in Fig. 9, which corresponds to the DC voltage, u_{dc} for all models. After model order reduction is conducted in the RSC, the Simplified-1 model structure, enclosed by dashed red lines in Fig. 9, shows missing states on the rotor side box corresponding to the machine-side AC current dynamics, and a new state appears on

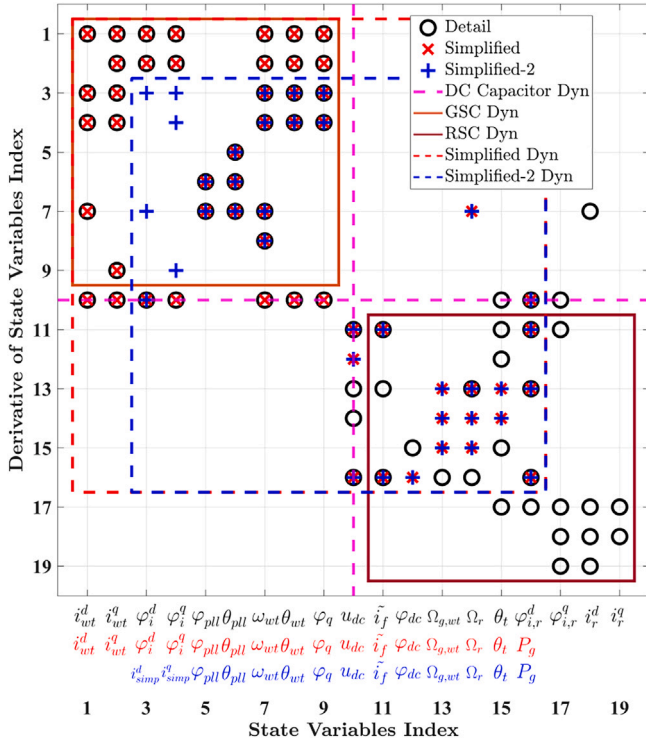


Fig. 9. Internal structure of the GFM WT state matrix.

the 16th corresponding to P_g . Nonetheless, the network side perturbation is propagated through the 10th state, i.e., the DC voltage, thus capturing rotor side dynamics. This is in agreement with the results of Fig. 5, which show a broad range of matching dynamics between these models. On the other hand, the rotor side dynamics would not be captured from a grid side perturbation for a GFL WT depending on the converter modulation strategy [22]. To help further understanding of the internal structure analysis for readers, a similar internal structure analysis of the GFL WT case and its implications on the interaction studies can be found in the Appendix. As demonstrated, this internal structure analysis helps to understand the path of physical interaction propagation between variables, thus providing more insights into the implications and justification of the different reduced order models.

3.4. Time-domain comparison of reduced order models

This subsection demonstrates the effectiveness of model order reduction with a single WT case in the EMT simulation environment. The effectiveness validation in the time domain is twofold. The first point is its accuracy in EMT simulation results. Second is the required computational power, which can be determined from the time spent on the EMT simulation. The CPU processor and software used in this paper for the validation of computational efficiency are the 13th Gen Intel(R) Core(TM) i7-1360 P(2.20 GHz) with 32GB of RAM and MATLAB.

The EMT test results for each model case are presented in Fig. 10 and Table 2. Fig. 10 shows almost matched responses between model types with respect to the disturbances of wind velocity V_w and connected grid side voltage magnitude $|u_g|$ at 30 and 100 sec, respectively. However, the latter response shows different initial transients, which correspond to the high-frequency dynamics. This re-confirms the analysis results of Fig. 5, highlighting the impacts of the current controller skipping from the model of Simplified-1 to the model of Simplified-2. Therefore, they show different high-frequency regions' dynamics as they have different current regulation capabilities, i.e., Simplified-2 has ideal current regulation, while others do not. Meanwhile, even though Fig. 8 also shows

Table 2

Computation time comparison of WT EMT simulation for each model type.

Model type	Computation time t_{com} (t_{sim} : 150 sec, Δt_{eml} : 10usec)	Relative time t_{com}/t_{sim}
Detailed	621.72 sec	4.1448 sec
Simplified-1	472.27 sec	3.1484 sec
Simplified-2	168.32 sec	1.1221 sec

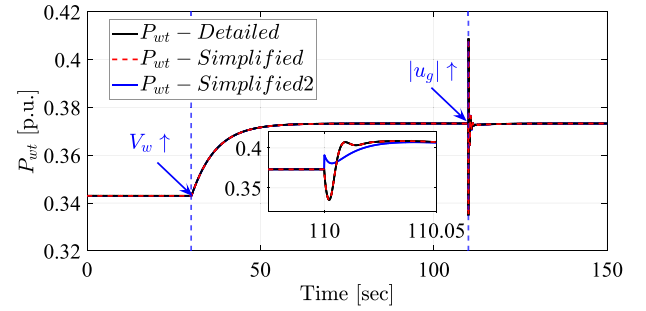


Fig. 10. Comparison of EMT time-domain simulation results. At 30 sec, 0.02 p.u. of V_w is increased. At 110 sec, 0.05 p.u. of grid side voltage magnitude $|u_g|$ is increased.

differences of magnitudes in the high frequency region, it has a negligible magnitude, thus there are no meaningful differences in the time domain results at 30 sec.

For the computational efficiency comparison with a single WT case connected to the grid side voltage source u_g in Table 2, it shows a gradual computational improvement of its efficiency in EMT simulation as the model order is reduced. Given the same simulation environment of total simulation time t_{sim} : 150 sec and time step used for EMT Δt_{eml} : 10 usec, they show different total spent time for simulation t_{com} . To effectively show its efficiency, the relative time t_{com}/t_{sim} is also calculated and presented. However, these advantages from the model order reduction only hold as long as the standalone stability of WT model is ensured. Therefore, this model order reduction approach is useful for the study of interaction with the interconnected system; however, it is not appropriate for the WT design study focusing on its internal stability to make a stable WT itself.

4. Inter-area oscillation analysis with GFM WT

In this section, small-signal stability analyses are carried out to identify potential risks associated with inter-area modes and GFM WTs.

4.1. Test system description

The system under study is a modified two-area system based on that in Ref. [4]. The system configuration is presented in Fig. 11, where one SG at each area is replaced by a GFM WT. Therefore, instead of the four electromechanical swing variables in the original two-area system [4], the modified system shows eight swing variables: two related to the SGs, two associated to the VSM controls, two linked to the PMSGs, and two from the WT drive-trains. For up-scaling of the GFM WTs capacity to represent a wind farm, the aggregated current value together with the total power capacity is used for the WT base while the voltage level is maintained. This approach neglects inter-farm interactions but is sufficient to capture farm-to-grid dynamics. The adopted system parameters, unless otherwise stated, are listed in Table C.4 of Appendix. Here, $H_{vsm,1}$ and $H_{vsm,2}$ are 6.5 s and 6.175 s, respectively. The constructed simplified analytical system model in Fig. 11 is validated via detailed EMT simulations. A step of 0.1 % on one AVR voltage reference is applied after 1 s and the speed deviations are shown in Fig. 12. The EMT variables

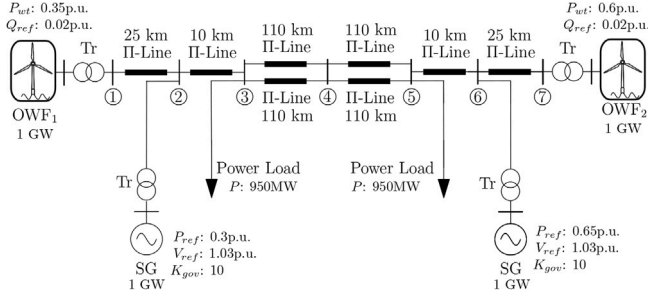


Fig. 11. Modified Two Area System with GFM WTs.

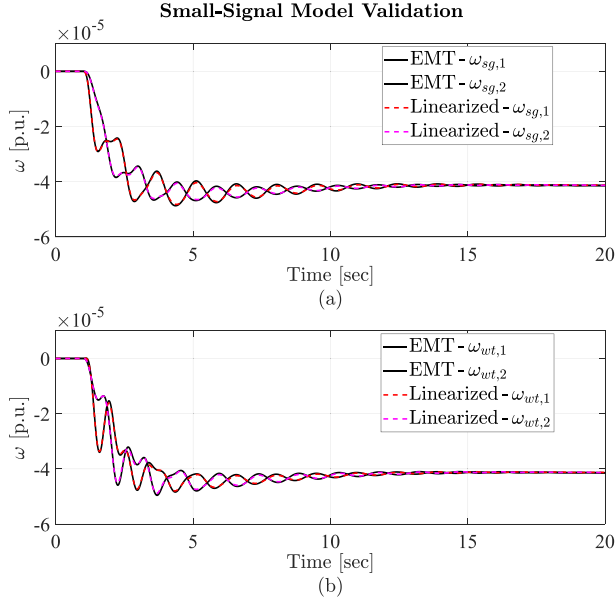
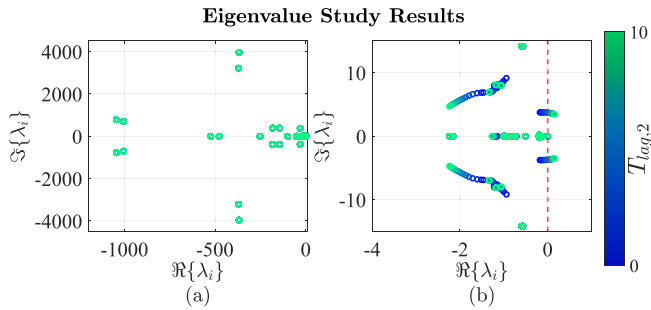


Fig. 12. Linearized model and EMT simulation validation.

Fig. 13. Eigenvalue study results under variation of $T_{lag,2}$. (a) Eigenvalues of entire system, (b) around inter-area oscillation mode frequency.

in black show a clear overlap with the linearized model in dashed lines, thus validating the developed model. These results support the usage of the simplified model for the study, as the simplified model well captures the slow dynamics of inter-area oscillations and drive train modes compared to the detailed model case.

4.2. Small-signal stability analysis

Firstly, to confirm whether similar inter-area instability can occur due to the AVR, a small-signal stability analysis is carried out varying the AVR parameter of SG2 $T_{lag,2}$, which is one of the well-known root

Table 3

Computation time comparison of modified two area system for each model case.

Used entire system model order & Type of GFM WT model and type	Computation time (Step size of $T_{lag,2}$: 0.05)
96th order with <i>Detailed</i> GFM WTs	567.4673 sec
90th order with <i>Simplified-1</i> GFM WTs	521.1799 sec
86th order with <i>Simplified-2</i> GFM WTs	504.5811 sec

EMT Simulation Results

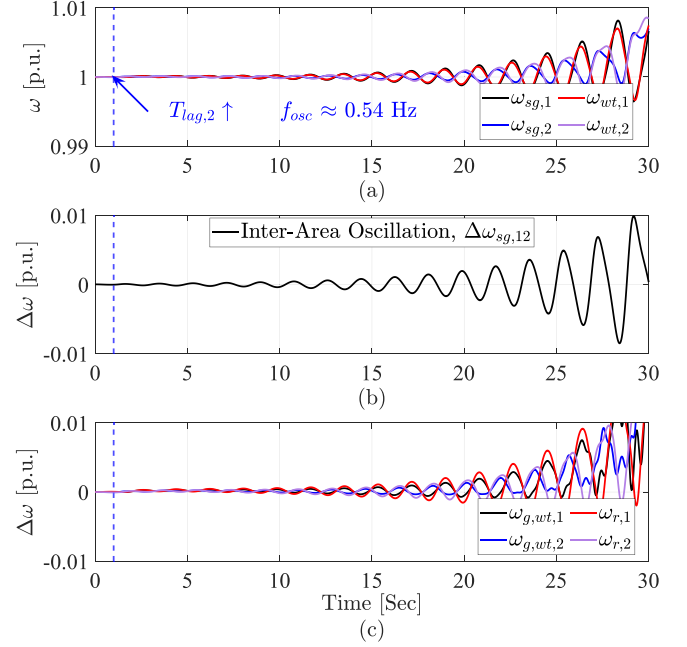


Fig. 14. Time domain simulation results of inter-area oscillation instability. (a) VSM and SG angular velocities, (b) inter-area oscillation, (c) drive train angular velocities variation.

causes of reduced inter-area mode damping [4]. The eigenvalue trajectory for this parameter sweep is presented in Fig. 13. As an eigenvalue around 3.4 rad/s moves to the right-half plane (RHP) for higher AVR lag time constant, it can be concluded that inter-area oscillatory instability can still occur with GFM WT in the system. This is because GFM control dynamics are trying to emulate SG's dynamic behavior; however, it highly depends on control parameters.

To validate the advantages of model order reduction, the same eigenvalue studies in Fig. 13 with different model types are investigated. Its computation time comparison results are provided in Table 3. The results show that the reduced order model brings simplicity and computational efficiency while maintaining the analysis accuracy of the low-frequency study. This demonstrated computational effectiveness is expected to increase as the size of the studied system, the number of WTs, and the step size of parameter investigation. Also, it holds for the time-domain simulations.

These analysis results are validated against non-linear EMT simulation and presented in Fig. 14. It reconfirms the effectiveness of reduced order model as EMT results are matched with eigenvalue studies by using the simplified model in Fig. 13. The 0.54 Hz inter-area oscillation instability is triggered when increasing $T_{lag,2}$ after 1 s, which confirms the small-signal stability assessment. In addition, the angular velocities of the SG, ω_{sg} , and the VSM of the GFM WT, ω_{wt} , swing against each other from each area, while those of the WT drive train ω_r and ω_l swing against each other from each area. This result implies that there is a risk of GFM WT drive-train damage due to inter-area mode

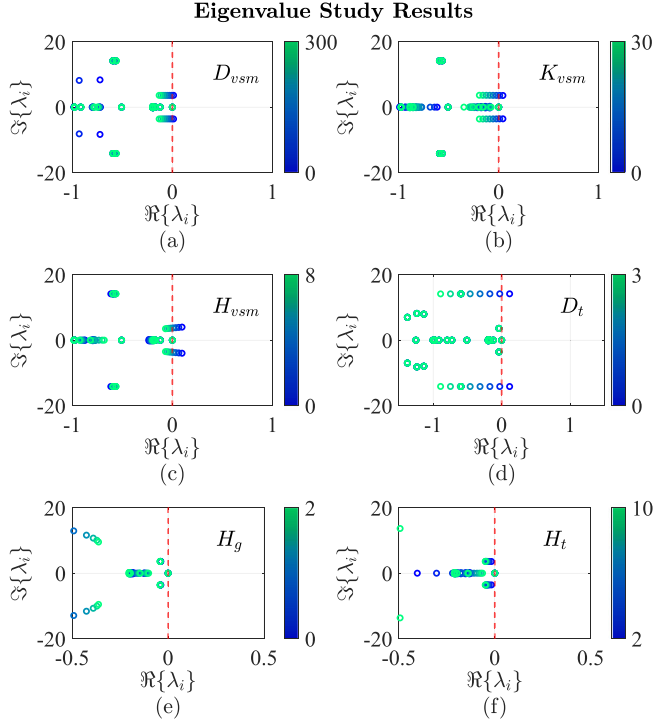


Fig. 15. Eigenvalue study results under variation of GFM WT parameters sweep with $T_{lag,2} = 7$. (a) Damping coefficient of VSM, (b) Droop gain of VSM, (c) Inertia constant of VSM, (d) Damping coefficient of drive train, (e) inertia constant of PMSG, (f) inertia constant of drive train.

interactions, even though the drive-train modes are not contributing to the instability. Subsequently, to investigate the potential (de)stabilizing of GFM WT in the inter-area modes, different GFM WT parameters are swept as the eigenvalue analysis is carried out leading to the eigenvalue trajectories in Fig. 15. The results show that VSM parameters significantly affect the inter-area modes. Depending on D_{vsm} and K_{vsm} , the damping of inter-area oscillation can be substantially improved. While the primary frequency response associated to K_{vsm} could lead to a certain amount of power reserve and is coordinated with the transmission system operator's requirements, the VSM damping D_{vsm} is a pure control parameter which can be more freely chosen, and thus a high value should be used to prevent inter-area modes instability with GFM WTs. Furthermore, the virtual inertia H_{vsm} directly affects the inter-area mode considerably. Even though it appears helpful to mitigate the inter-area oscillation concerns, it still requires extra caution since the admissible range of virtual inertia is also affected by its dynamic performance, and a high value of virtual inertia does not necessarily always result in better stability margins [23].

Meanwhile, the inertia constant of the drive train H_t shows less effect on this inter-area mode. This is because the drive train modes' natural mode frequency $\omega_{t,osc}$ from Eq. (17) with the considered parameters is sufficiently decoupled from the inter-area mode. Therefore, the system inter-area mode should be taken into account together with the drive-train oscillation mode based on Eq. (17) in order to avoid a negative impact from the GFM WT on inter-area oscillations.

The drive train mode is highly influenced by its damping parameter D_t , and thus the impact of this parameter on the GFM WT interaction with entire system can be evaluated via frequency-domain analysis. Intuitively, if wind speed V_w changes occur, these would propagate through the drive-train into the WT and the rest of the power system.

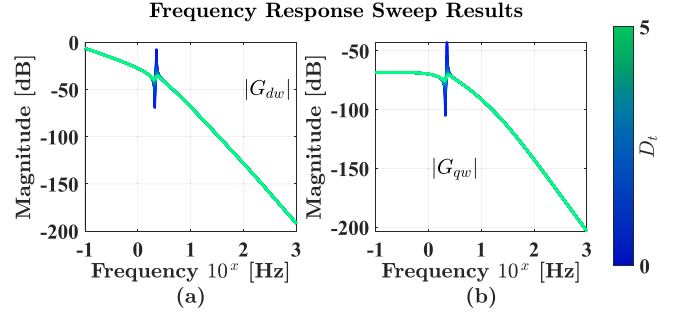


Fig. 16. Frequency response of i_{wt} with respect to V_w sweep results with D_t .

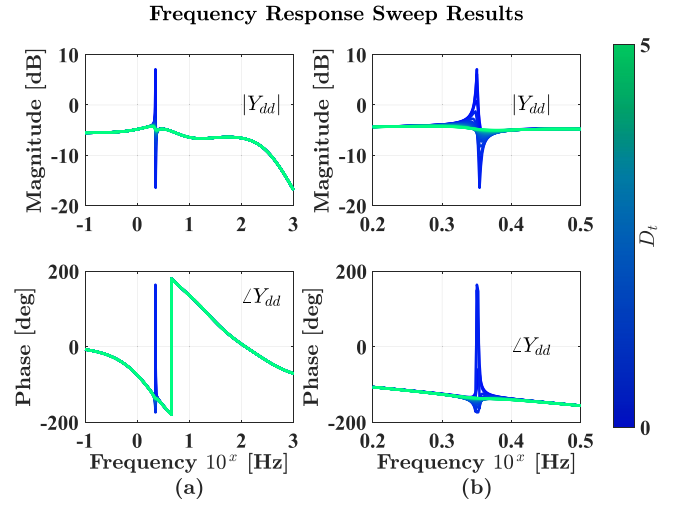


Fig. 17. Frequency response of i_{wt} with respect to u_{wt} sweep results with D_t .

Hence, the transfer functions between wind speed and WT grid-side current magnitudes are examined for a sweep of the drive-train damping in Fig. 16. The drive train creates a sharper resonance with lower damping parameter D_t , while the resonant frequency is not affected as specified in Eq. (17).

If the WT grid-side voltage u_{wt} is taken as input, the admittance can be evaluated for the same parametric sweep. Fig. 17 shows the Y_{dd} element for compactness presenting similar effects as those considering wind speed as input in Fig. 16: the influence of between the network-side and the drive-train variables can be decreased for higher D_t . As a result, a high value of D_t is also helpful for mitigating the impact of GFM WT's drive-train on the interaction with the interconnecting system at the expense of higher mechanical losses.

These findings are aligned with the eigenvalue results in Fig. 15. Therefore, having proper drive train damping D_t and/or high VSM damping D_{vsm} is important not only for stability purposes but also for mitigating the negative impact of system disturbances on the GFM WT.

Similarly, the effect of the VSM parameters can be captured by the GFM WT admittance. To highlight the impact of individual VSM parameters, here the base parameters of VSM used are K_{vsm} , D_{vsm} , and H_{vsm} are 0, 0, and 6.5, respectively. It should be noted that these parameter settings are only for understanding their impacts on observed inter-area oscillation. As it is more related to GFM control, frequency response with respect to u_{wt} is examined. In Fig. 18, the Y_{dq}

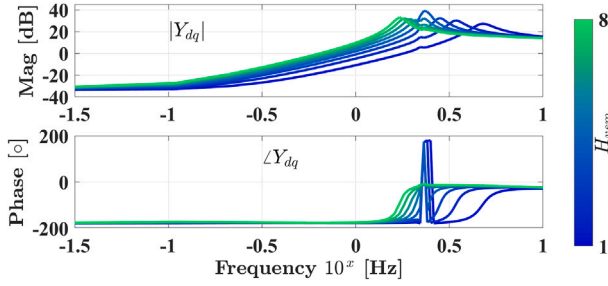


Fig. 18. Frequency response of i_{wt} with respect to u_{wt} sweep results with H_{vsm} .

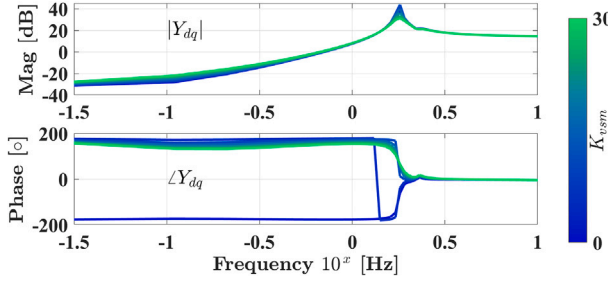


Fig. 19. Frequency response of i_{wt} with respect to u_{wt} sweep results with K_{vsm} .

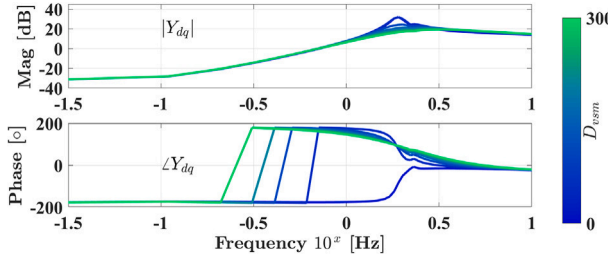


Fig. 20. Frequency response of i_{wt} with respect to u_{wt} sweep results with D_{vsm} .

admittance element is shown for different VSM inertia constant values. The increased admittance magnitude for higher inertia constants corresponds to a larger GFM WT response to electrical disturbances. As expected, it shifts the frequency of the peak in magnitude, i.e., it is highly related to the response speed (shift of swing mode) rather than damping. Meanwhile, the impacts of K_{vsm} and D_{vsm} are shown in Fig. 19 and Fig. 20. Both show similar damping effects on swing mode by mitigating the magnitude of the responses. While H_{vsm} shows the increased magnitude with in-phase, others show decreased magnitude with out-of-phase responses around the concerned frequency region. Therefore, there are different mechanisms to stabilize the inter-area oscillation between H_{vsm} with K_{vsm} and D_{vsm} . As a result, it can be understood that H_{vsm} helps avoid the inter-area oscillation by increasing synchronizing power. This is well aligned with the observation from Fig. 13.

4.3. Time domain simulations

In this section, previous results are verified by EMT simulations. In particular, the effect of the VSC inertia and drive-train damping obtained by means of the reduced order models are confirmed with respect to the complete non-linear model.

The time-domain simulation in Fig. 21 shows how decreasing the VSM inertia of the GFM WT2 after 1 s can lead to an unstable inter-area

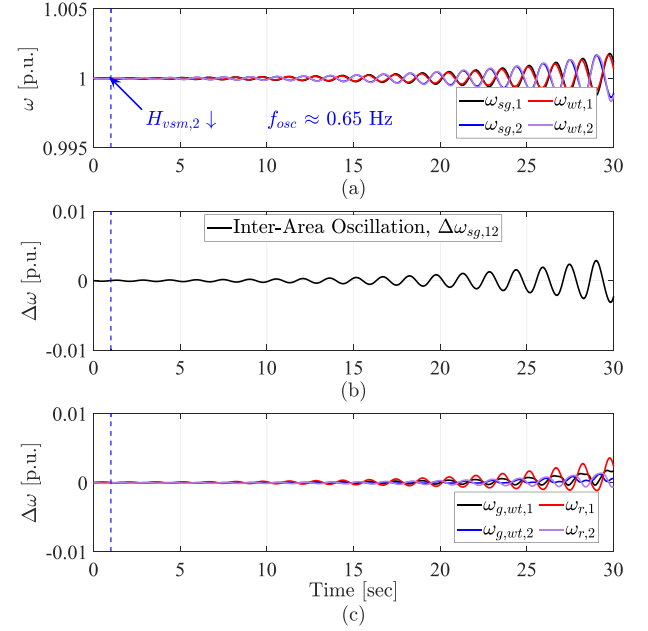


Fig. 21. Time domain simulation results with the disturbance of $H_{vsm,2}$ step down to 3 s. (a) VSM and SG angular velocities, (b) inter-area oscillation, (c) drive train angular velocities variation.

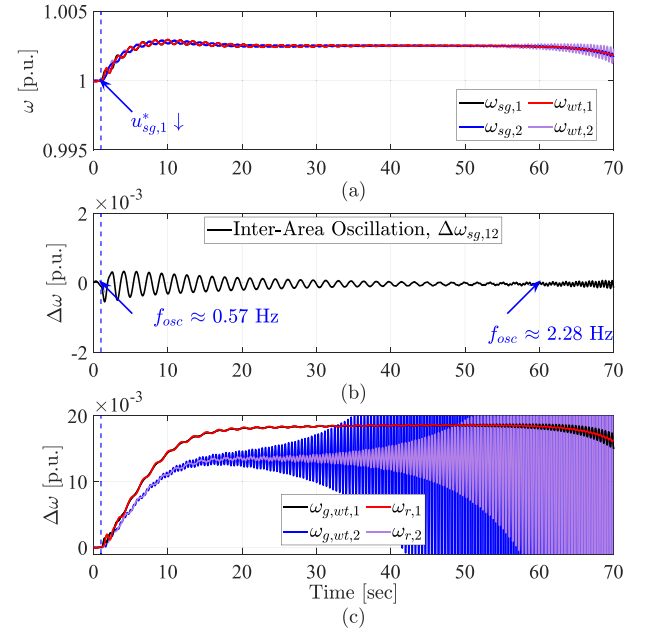


Fig. 22. Time domain simulation results after a $u_{sg,1}^*$ step down of 0.03 p.u. for $D_l = 0.1$ p.u. (a) VSM and SG angular velocities, (b) inter-area oscillation, (c) drive train angular velocities variation.

mode. Fig. 21 (a) indicates that the SG and GFM WT at area two are oscillating together against the SG and GFM WT at area one. The difference between the SG speed at each area in Fig. 21 (b) contains the clearly unstable oscillatory inter-area mode around 0.65 Hz. Lastly, the drive-train variables in Fig. 21 (c) also manifest the unstable oscillation, which highlights the importance of GFM WT to positively contributing to the inter-area modes damping so as to avoid mechanical stress and damage.

The negative effect of the drive-train on the system stability can be validated by considering the case with low drive-train damping $D_i = 0.1$ p.u. and perturbing the system from its equilibrium point to assess whether it corresponds to an stable or unstable operating point. The EMT simulation results after a -0.03 p.u. step on the AVR voltage reference of SG1, $u_{sg,1}^*$, is applied at 1 s are shown in Fig. 22. The low-damped but stable 0.57 Hz inter-area oscillation can be observed in all variables, while the GFM WT drive-train speeds show a growing oscillation of 2.28 Hz. This frequency corresponds to the drive-train mode visible as a peak in Fig. 17, highlighting the importance of a correct damping of the internal GFM WT natural modes.

5. Conclusion

This paper addresses the concern of inter-area oscillation in the presence of GFM-WTs. To this end, a GFM-WT small-signal model is built and validated with respect to an EMT model via frequency scans, which is used as a benchmark for examining the validity of reduced-order models. The fidelity of two reduced order models is assessed through a comprehensive investigation including modal analysis, frequency response, and internal model structure comparisons. These approaches bring a better understanding of the impact of modeling fidelity on the study of interaction assessment. The results indicate that the rotor-side variables can impact the grid-side, and that rotor-side AC current dynamics can be simplified without significant loss of accuracy. Further model order reduction in the grid-side converter leads to accurate GFM-WT modeling below approximately 10 Hz, and thus adequate for the efficient study of inter-area phenomena. In addition, a modified two-area power system with high penetration of GFM-WTs and detailed SG models is used to evaluate the impact of several WT parameters, such as the VSM damping and inertia, on inter-area modes. Frequency response and eigenvalues trajectories are presented to examine the system dynamics under parametric sweeps. GFM-WT control parameters, such as the VSM damping and frequency droop, have major impact on the inter-area modes and could lead to unstable inter-area oscillations. Predominantly, higher VSM inertia and damping are found to increase the damping of inter-area modes, while low drive-train damping can lead to GFM-WT internal instability. These findings, validated via non-linear EMT simulations, can guide analysts to select more appropriate GFM-WT models as well as system operators to better understand the implications of this technology on system-wide interactions. For future works, a detailed representation of the wind farm with several individual wind turbines will be considered together with the slower farm-level control, as there are potential interactions between turbines and detailed related dynamics such as wake effects.

CRedit authorship contribution statement

Dongyeong Lee: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Francisco Javier Cifuentes Garcia:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jef Beerten:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Synchronous generator modeling

The machine electromagnetic equations can be expressed in state-space form and in per unit using $\mathbf{p}$ to denote the time-derivative operator and other bold symbols to denote vectors and matrices as

$$\mathbf{p} \begin{bmatrix} i^d \\ i_f \\ i_1^d \end{bmatrix} = \omega_b \mathbf{L}_D^{-1} \begin{bmatrix} u^d + R_s i^d - \omega_{sg} \psi^q \\ u_f - R_f i_f^d \\ -R_1^d i_1^d \end{bmatrix} \quad (\text{A.1})$$

$$\mathbf{p} \begin{bmatrix} i^q \\ i_1^q \\ i_2^q \end{bmatrix} = \omega_b \mathbf{L}_Q^{-1} \begin{bmatrix} u^q + R_s i^q + \omega_{sg} \psi^d \\ -R_1^q i_1^q \\ -R_2^q i_2^q \end{bmatrix} \quad (\text{A.2})$$

$$\boldsymbol{\psi} = - \begin{bmatrix} L_m^d + L_s & 0 \\ 0 & L_m^q + L_s \end{bmatrix} \begin{bmatrix} i^d \\ i^q \end{bmatrix} + \begin{bmatrix} L_m^d & 0 \\ 0 & L_m^q \end{bmatrix} \begin{bmatrix} i_f + i_1^d \\ i_1^q + i_2^q \end{bmatrix} \quad (\text{A.3})$$

$$\mathbf{L}_D = \begin{bmatrix} -(L_m^d + L_s) & L_m^d & L_m^d \\ -L_m^d & L_f + L_m^d & L_m^d \\ -L_m^d & L_f & L_1^d + L_m^d \end{bmatrix} \quad (\text{A.4})$$

$$\mathbf{L}_Q = \begin{bmatrix} -(L_m^q + L_s) & L_m^q & L_m^q \\ -L_m^q & L_1^q + L_m^q & L_m^q \\ -L_m^q & L_m^q & L_2^q + L_m^q \end{bmatrix} \quad (\text{A.5})$$

where ω_{sg} is the rotor angular speed in per unit, $\omega_b = 2\pi f_b$ is the nominal grid angular speed in rad/s with $f_b = 60$ Hz, u^d, u^q are the instantaneous stator phase voltages, i^d, i^q are the instantaneous stator currents, $\boldsymbol{\psi} = [\psi^d, \psi^q]^T$ are the stator flux linkages, u_f is the field voltage, i_f is the field current, R_f is the field resistance, L_f is the field inductance, i_k^d, i_k^q are the damper currents with k being the number of the damper, R_k^d, R_k^q and L_k^d, L_k^q are the k -th damper windings resistances and inductances for the d and q axes, L_m^d, L_m^q are the rotor-stator magnetizing inductances and L_s, R_s are the armature-winding leakage inductance and resistance, respectively which can also incorporate the transformer leakage impedances, L_t, R_t . The equation of motion is given by (A.6) where T_e is the electrical SG torque in per unit and H_{sg} is the machine inertia constant in seconds.

$$\mathbf{p}\omega_{sg} = \frac{T_m - T_e}{2H_{sg}}, \quad T_e = \psi^q i^d - \psi^d i^q \quad (\text{A.6})$$

The AVR model and field voltage can be described by (A.7)

$$G_{AVR}(s) = \frac{u_f}{u_{sg}^* - u_{sg}} = \frac{K_A}{1 + sT_{v,avr}} \frac{1 + sT_{lead}}{1 + sT_{lag}} \frac{L_a^d}{R_f} \quad (\text{A.7})$$

where $T_{v,avr}, T_{lead}, T_{lag}$ are time constants and K_A is the static gain. The turbine and governor dynamics can be represented in by the following transfer functions, respectively

$$\frac{\Delta T_m}{\Delta T^*} = \frac{1}{1 + sT_{CH}} \left(F_{CH} + \frac{F_{RH}}{1 + sT_{RH}} + \frac{1}{1 + sT_{RH}} \frac{F_{CO}}{1 + sT_{CO}} \right)$$

$$G_{gov}(s) = \frac{\Delta T^*}{\omega_{ac}^* - \omega_{sg}} = \frac{K_{gov}}{1 + sT_{w,m}} \frac{1}{1 + sT_G}$$

where T_{CH}, T_{RH}, T_{CO} are the time constants of each of the steam turbine components, and F_{CH}, F_{RH}, F_{CO} their torque fractions, and $T_{w,m}, T_G$ are the measurement and actuator time constants and proportional gain K_{gov} of the governor.

Appendix B. GSC of GFM WT modeling

GSC control of GFM WT in addition to GFM loop of Eq. (2) and Eq. (3) [14]:

$$e_q = Q_{wt}^* - Q_{wt} \quad (B.1)$$

where Q_{wt}^* is reactive power reference of WT and is set to be zero in this paper. Q_{wt} is measured reactive power. This error signal goes to reactive power PI controller together with internal voltage magnitude reference u_{int}^* as

$$u_{wt}^{d,*} = K_{p,q} e_q + \varphi_q + u_{int}^* \quad (B.2)$$

$$u_{wt}^{q,*} = 0 \quad (B.3)$$

where u_{wt}^* is internal voltage reference.

$$e_{u,wt}^d = u_{wt}^{d,*} - u_{wt}^d \quad (B.4)$$

$$e_{u,wt}^q = u_{wt}^{q,*} - u_{wt}^q \quad (B.5)$$

where $e_{u,wt}$ is error signals of internal voltage signals. u_{wt} is measured PCC voltages of GSC. This error signals goes to virtual impedance control loop as

$$i_{wt}^{d,*} = |Y_v|^2 (e_{u,wt}^d r_v + e_{u,wt}^q \omega_{wt} l_v) \quad (B.6)$$

$$i_{wt}^{q,*} = |Y_v|^2 (e_{u,wt}^q r_v - e_{u,wt}^d \omega_{wt} l_v) \quad (B.7)$$

$$|Y_v| = \frac{1}{\sqrt{r_v^2 + l_v^2}} \quad (B.8)$$

where r_v and l_v are virtual resistance and inductance, respectively. $|Y_v|$ denotes the magnitude of the virtual admittance. These current control references i_{wt}^* goes to current controller as

$$u_{c,wt}^d = G_{i,gs}^d (i_{wt}^{d,*} - i_{wt}^d) - \omega_{wt} L_c i_{wt}^q + u_{wt}^d \quad (B.9)$$

$$u_{c,wt}^q = G_{i,gs}^q (i_{wt}^{q,*} - i_{wt}^q) + \omega_{wt} L_c i_{wt}^d + u_{wt}^q \quad (B.10)$$

where $G_{i,gs}$ denotes the transfer function of the current PI controller. L_c is filter inductance of GSC.

Appendix C. The used system parameters

Network parameters can be found in Ref. [4]. Table C.4 shows other parameters.

Table C.4

Parameters of studied two area system.

Parameter list			
Parameters	Value [p.u.]	Parameters	Value [p.u.]
System			
S_b	1000 MW	F_{LP}, F_{LP}	0.4, 0.3
u_b	230 kV	WT	
ω_b	60 Hz	L_c, R_c	0.15, 0.02
Load	950 MW	H_{usm}, L_v	3.0, 0.2
SG		K_{usm}, R_v	20, 0.02
L_i, R_i	0.15, 0.015	D_{usm}, K_{usm}	100, 10
L_m^d, L_1^d	1.66, 0.1713	K_i, D_i	278.61, 1.99
L_f, R_1^d	0.165, 0.02	H_i, H_g	5, 0.8
R_f, L_m^q	0.002, 1.61	T_c, r_g	0.5sec, 0.017
L_1^q, L_2^q	0.725, 0.125	L_g^d, L_g^q	0.55, 1.11
R_1^q, R_2^q	0.005, 0.01	$K_{p,dc}, K_{i,dc}$	1.94, 18.95
L_i, R_i	0.15, 0.0015	$K_{p,q}, K_{i,q}$	0.016, 31.41
$K_{avr}, T_{v,avr}$	100, 0.05sec	$K_{p,i}, K_{i,i}$	0.9, 37.69
T_{lag}, T_{lead}	5sec, 1sec	ψ_M	1.0
$T_{w,m}, T_G$	0.1sec, 0.2sec	$T_{i,m}$	0.0001sec
K_{gov}, H_{sg}	10, 6.5sec	$T_{i,m}$	0.002sec
T_{CH}, T_{RH}	2sec, 5sec	$K_{p,pil}, K_{i,pil}$	21.58, 232.99
T_{CO}, F_{HP}	1sec, 0.3	T_g, T_{cc}	0.003sec, 0.003sec

Appendix D. Internal structure analysis of GFL WT case

Depending on the modulation strategy, converter voltage outputs can be expressed as follows:

$$u_c = m^* u_{dc}, \quad \begin{cases} m^* = u_c^* / u_{dc} & \text{Compensated} \\ m^* = u_c^* / u_{dc}^0 & \text{Uncompensated} \end{cases} \quad (D.1)$$

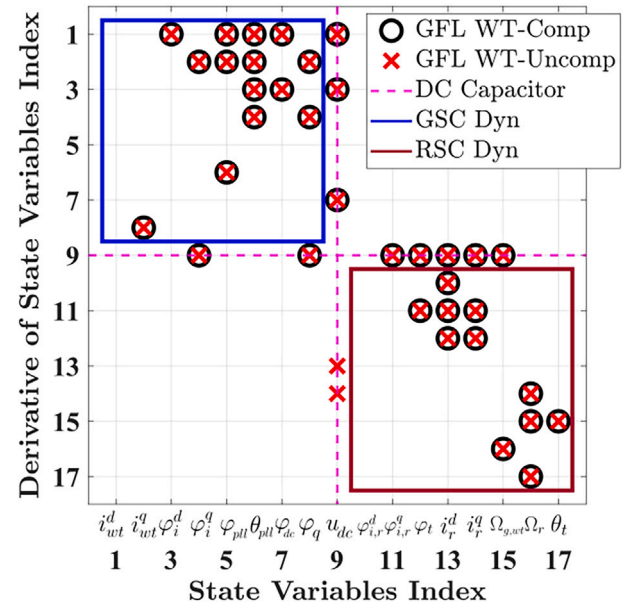


Fig. D.23. Internal structure of the GFL WT state matrix.

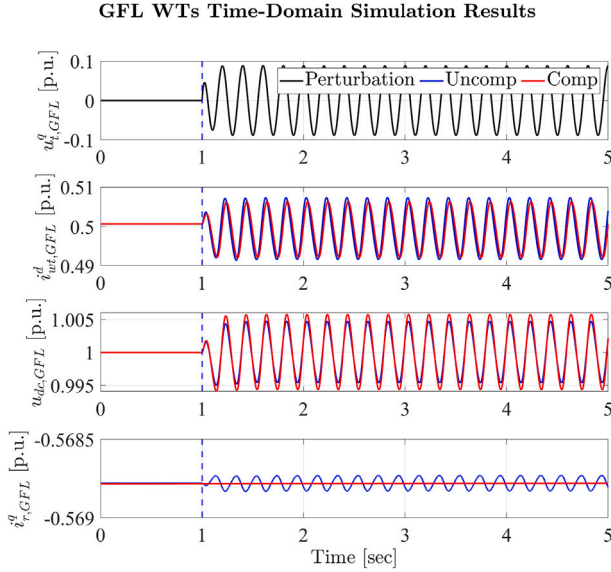


Fig. D.24. Comparison results of the grid side perturbation responses between different modulation strategies.

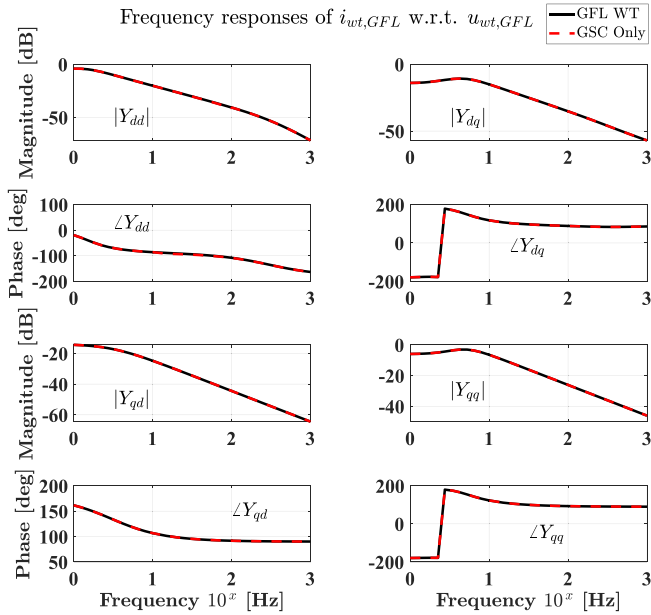


Fig. D.25. Frequency response comparison between GFL WT model and only GSC with respect to grid side voltages.

where u_c denotes converter output terminal voltages after modulation. m^* modulation reference signal and $u_{dc,wt}^0$ denotes nominal DC voltage. Here, u_c^* is the converter control output of the current controller. From (D.1), variation of u_{dc} will be canceled out for its u_c for the Compensated strategy case. Therefore, the impact of u_{dc} perturbation on grid-side interaction cannot be seen from the grid-side response of the converter. i.e., it can be observed in the Uncompensated strategy case.

Utilizing this phenomenon, GFL WT case is demonstrated with the internal structure analysis to further help the understanding of the impact of model order reduction for the study of system-wide interactions. The used GFL WT model with state variables are $x_{GFL} = [i_{wt}^d, i_{wt}^q, \phi_i^d, \phi_i^q, \phi_{pll}, \theta_{pll}, \phi_{dc}, \phi_q, u_{dc}, \phi_{i,r}^d, \phi_{i,r}^q, \phi_r, i_r^d, i_r^q, \Omega_{g,wt}, \Omega_r, \theta_r]^T \in \mathbb{R}^{17}$, where subscript ϕ_r is turbine torque controller variable in RSC

of GFL WT. Depending on the modulation strategy, internal structure analysis results are presented in Fig. D.23. The results show that DC Capacitor dynamics is an interaction propagation path in the same manner with Fig. 9. However, in the case of the Compensated strategy, there is no path of interaction propagation from the GSC dynamics to the RSC dynamics (left-below square based on DC Capacitor dynamics). In contrast, the uncompensated strategy case shows an interaction propagation path through DC voltage dynamics. These results imply that rotor-side dynamics can be neglected when focusing on the network-side interaction as there is no propagation path from grid-side perturbation to rotor-side dynamics.

To validate the internal structure analysis results, time-domain simulations are carried out to observe the impact of the interaction propagation path, and the results are shown in Fig. D.24. The results show that the grid side voltage perturbation signal u_i^q results in the response of GSC current i_{wt}^d and DC voltage u_{dc} . However, the rotor side current i_r^q shows prominently different results between modulation strategies. As the above results show, rotor side dynamics do not respond to the network side perturbation in the compensated strategy case. In addition, u_{dc} and i_{wt}^d also show slightly different responses in terms of magnitude and phase shift between strategies due to the reflection of the rotor side dynamics with respect to network side perturbation. As a result, rotor side dynamics interact with the network for the uncompensated case. Based on the analysis results, the compensated strategy case can achieve model order reduction by neglecting rotor side dynamics, and its results are validated through frequency responses in Fig. D.25.

Data availability

The data that has been used is confidential.

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